not only replace manned reconnaissance aircraft in many roles but also provide safer, more efficient, and cost-effective alternatives for sustained operations.

In essence, the RQ-1 Predator (1995) symbolizes the dawn of networked, precision-oriented drone warfare. Its legacy extends beyond the battlefield; it established the technological and strategic foundation for successors such as the MQ-9 Reaper and RQ-4 Global Hawk, and it permanently altered the role of UAVs in both military and intelligence operations worldwide.

*Key specifications*:

- Type: Medium-altitude long-endurance (MALE) UAV
- Developer: General Atomics Aeronautical Systems, United States
- First Flight: 1995
- Wingspan: 17.0 m
- Endurance: Up to 24 hours
- Maximum Altitude: 25,000 ft
- Propulsion: Rotax 914 piston engine, 115 hp
- Payloads: EO/IR sensors, SAR radar, laser designator, Hellfire missiles (MQ-1 variant)
- Control system: Satellite (Ku-band) and C-band line-of-sight datalink
- Primary Roles: ISR, target acquisition, and precision strike

Figure 4.2 illustrates the structural layout and key operational components of the RQ-1 Predator, one of the most significant UAVs of the third generation.